

JARED DUONG

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📁 [Portfolio](#)

Experience

Software Integration Developer

Jan 2026 – Mar 2026

Appleby North Paint Inc. – Ripen Level UP Project

Remote

- Built a **REST-based integration** between **QuickBooks** and **Decor Fusion** to automate sales and inventory synchronization.
- Mapped invoice, transaction, and inventory fields across accounting and point-of-sale systems to reduce manual entry.
- Documented integration setup, field mappings, and troubleshooting procedures to support ongoing maintenance.

Production Student

May 2025 – Jun 2025

Hartmann Canada

Brantford, ON

- Analyzed production data and monitored **automated equipment** to identify recurring process and equipment faults.
- Collaborated with operators and engineers to **troubleshoot mechanical and production faults**.
- Documented recurring faults and process issues to guide troubleshooting and support production improvements.

Projects

16-Bit CPU & FPGA Digital Systems | VHDL, Vivado, ModelSim

Jan 2026 – Mar 2026

- Co-developed a **16-bit CPU in VHDL**, integrating an ALU, register file, control unit, and instruction/data memory.
- Designed a **16-bit ripple-carry ALU** from modular 1-bit slices and verified arithmetic and logic operations in simulation.
- Implemented and tested digital systems on a **Nexys A7 FPGA**, including clock division and multiplexed display control.
- Used testbenches and waveform analysis to verify ALU outputs, register updates, and memory operations.

Reverse Engineering & CAD Reconstruction | SOLIDWORKS

Jan 2026 – Feb 2026

- Collaborated on reverse engineering a **12-in-1 Solar & Hydraulic Construction Kit** in **SOLIDWORKS**, recreating component geometry and assemblies.
- Reconstructed component placement and interfaces to reproduce the kit's mechanical relationships.
- Analyzed part interactions and assembly constraints to understand mechanism function and movement.

4x4x4 LED Cube with Custom PCB | Arduino, PCB Design, Digital Logic

Mar 2025 – Apr 2025

- Designed and soldered a **64-LED cube** using discrete logic ICs (**74HC04**, **SN74HC08**) and a custom-routed **PCB**.
- Programmed multiplexed LED control patterns using an **Arduino Nano** in **C/C++**.
- Implemented layered scanning and timing logic to control 3D LED addressing and layer activation.
- Designed and 3D-printed a structural base to house the PCB, Arduino, wiring, and control electronics.

Autonomous Wheelchair Prototype | Arduino, C++, SOLIDWORKS

Sep 2024 – Dec 2024

- Developed an **Arduino-powered autonomous wheelchair prototype** integrating sensors and motor control for navigation.
- Performed **center-of-mass and tipping analysis** to evaluate stability and guide mechanical design changes.
- Evaluated component costs and revised the prototype design to reduce expenses while maintaining functionality.

Education

University of Guelph

Sep 2024 – Apr 2029

Bachelor of Engineering in Computer Engineering (Co-op)

Guelph, ON

Technical Skills

Languages: C, C++, VHDL, Python, Java, SQL, C#

Embedded / Hardware: Arduino, FPGA (Nexys-3, Nexys A7), Digital Logic Design, Sensor Integration, Motor Control, PCB Design, Soldering, Circuit Analysis, Robotics

Tools: Vivado, ModelSim, MATLAB, Git & GitHub, Visual Studio Code

CAD / Design: SOLIDWORKS, Fusion 360, AutoCAD, 3D Printing

Certifications & Training: Certified SOLIDWORKS Associate (CSWA); Microsoft Applied Skills: Develop a Generative AI Chat App Using the Microsoft Foundry SDK; OpenAI Academy – Agents and Workflows, Applied AI Foundations

Engineering Involvement

University of Guelph Robotics Team – Mechanical Team Member

Sep 2024 – Present

University of Guelph

Guelph, ON

- Design and build mechanical components for competition robots using **SOLIDWORKS** and **AutoCAD**.
- Collaborate with electrical and software teammates to integrate sensors, controls, and mechanical systems.
- Prototype and test components, refine CAD designs, and troubleshoot assembly and integration issues.